

The vectors v_1 and v_2 are linearly dependent vectors for some value of a .

Options :

6406535120922. ✖ TRUE

6406535120923. ✔ FALSE

Question Number : 43 Question Id : 6406531529705 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

For $a = 0$, find the value of b such that the set $\{v_1, v_2, v_3\}$ is not a basis of $\mathbb{R}^3$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1.5

Corporate Finance

Section Id :	640653106140
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	26
Number of Questions to be attempted :	26
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1

Sub-Section Id :

640653245465

Question Shuffling Allowed :

No

Question Number : 44 Question Id : 6406531529706 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : CORPORATE FINANCE (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535120925. ✓ YES

6406535120926. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653245466

Question Shuffling Allowed :

Yes

Question Number : 45 Question Id : 6406531529707 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A zero-coupon bond with face value of Rs 5,000 matures in one year. Its price is Rs 4,500. What is the yield of the bond?

Options :

6406535120927. ✗ 8.42%

6406535120928. ✗ 10.00%

6406535120929. ✓ 11.11%

6406535120930. ✗ 9.00%

Question Number : 46 Question Id : 6406531529708 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

If you want to double your money at 5% annual rate of interest, how many years will it take?

Options :

6406535120931. ✓ $\log(2)/\log(1.05)$

6406535120932. ✗ $\log(1.05)/\log(2)$

6406535120933. ✖ $\log(2)/\log(0.05)$

6406535120934. ✖ $\log(0.05)/\log(2)$

Question Number : 47 Question Id : 6406531529709 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

You purchased an asset today for Rs 1,000. Suppose the asset earns an annual interest rate of 7% which is continuously compounded. What will be the value of the asset after 5 years (Rounded off to nearest integer)?

Options :

6406535120935. ✖ Rs 1350

6406535120936. ✖ Rs 1402

6406535120937. ✔ Rs 1419

6406535120938. ✖ Rs 1500

Question Number : 48 Question Id : 6406531529710 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which rule helps an entrepreneur with limited capital get back invested money fastest in high interest environments?

Options :

6406535120939. ✖ NPV

6406535120940. ✔ Payback

6406535120941. ✖ IRR

6406535120942. ✖ Profitability Index

Question Number : 49 Question Id : 6406531529711 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

If nominal GDP doubles, but real GDP remains unchanged, this means

Options :

6406535120943. ✔ Only prices increased.

6406535120944. ✖ Only quantity increased.

6406535120945. ✖ Both prices and quantity increased.

6406535120946. ✖ Neither prices nor quantity increased.

Question Number : 50 Question Id : 6406531529712 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

The PVC stock is trading at Rs 300 today. PVC is expected to pay a dividend of Rs 20 per share next year. If the growth rate of the PVC dividend is 3% every year and the appropriate rate of return is 10% per year, then according to the Gordon growth model

Options :

- 6406535120947. ✖ The PVC stock is underpriced, and an investor should buy PVC shares.
- 6406535120948. ✖ The PVC stock is overpriced, and an investor should buy PVC shares.
- 6406535120949. ✖ The PVC stock is underpriced, and an investor should sell PVC shares.
- 6406535120950. ✔ The PVC stock is overpriced, and an investor should sell PVC shares.

Question Number : 51 Question Id : 6406531529713 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A bond pays 200 units in 1 year and 200 units in 5 years. What is the (simple) duration of the bond?

Options :

- 6406535120951. ✖ 5 years
- 6406535120952. ✔ 3 years
- 6406535120953. ✖ 2.5 years
- 6406535120954. ✖ 1 year

Question Number : 52 Question Id : 6406531529714 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

In a 3-bond pool, each bond has 20% default probability, and these bonds are perfectly correlated to each other ($\rho = 1$). What is the probability that all three bonds default together (for a senior tranche)?

Options :

- 6406535120955. ✖ 0.008
- 6406535120956. ✖ 0.128
- 6406535120957. ✔ 0.200
- 6406535120958. ✖ 0.512

Question Number : 53 Question Id : 6406531529715 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

If expected inflation equals nominal interest rate, then real interest rate is

Options :

- 6406535120959. ✖ Positive
- 6406535120960. ✖ Negative

6406535120961. ✓ Zero

6406535120962. ✗ One

Question Number : 54 Question Id : 6406531529716 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An upward sloping yield curve shows:

Options :

6406535120963. ✗ Short-term rates are higher than long-term rates.

6406535120964. ✓ Short-term rates are lower than long-term rates.

6406535120965. ✗ Maturity is irrelevant.

6406535120966. ✗ It is one of the indications of upcoming recession.

Question Number : 55 Question Id : 6406531529717 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose there is a new project that demands an investment of Rs 10 lakh today and promises to pay off after two years. If the estimated internal rate of return of the project is 12%, then what amount is paid off on the payment date (after two years)?

Options :

6406535120967. ✗ Rs 12 lakh

6406535120968. ✗ Rs 12.400 lakh

6406535120969. ✓ Rs 12.544 lakh

6406535120970. ✗ Rs 12.838 lakh

Question Number : 56 Question Id : 6406531529718 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

During the cleaning of house, you find an old sword that belonged to your grandfather. Your grandfather purchased the sword 60 years ago for Rs 60. You go to the market and find out that the current value of the sword is Rs 9,000. What is the annualized rate of return that the sword has earned?

Options :

6406535120971. ✗ 60%

6406535120972. ✗ 10.87%

6406535120973. ✓ 8.71%

6406535120974. ✗ 2.50%

Question Number : 57 Question Id : 6406531529719 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

ABC Corporation is looking to invest in one new project, and they have three options: Project-A is expected to generate Rs 30 lakh after two years and Rs 40 lakh after 5 years; Project-B is expected to generate Rs 10 lakh each year for 6 years (beginning next year); Project-C is expected to generate Rs 100 lakh after 10 years. Each one of the project costs Rs 40 lakh today. The required rate of return on these projects is 10%. What is the profitability index of Project-B?

Options :

6406535120975. ✓ 1.09

6406535120976. ✗ 1.19

6406535120977. ✗ 1.88

6406535120978. ✗ 0.88

Question Number : 58 Question Id : 6406531529720 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

ABC Corporation is looking to invest in one new project, and they have three options: Project-A is expected to generate Rs 30 lakh after two years and Rs 40 lakh after 5 years; Project-B is expected to generate Rs 10 lakh each year for 6 years (beginning next year); Project-C is expected to generate Rs 100 lakh after 10 years. Each one of the project costs Rs 40 lakh today. The required rate of return on these projects is 10%. Which one of the projects should ABC Corporation choose according to NPV criterion?

Options :

6406535120979. ✓ Project-A

6406535120980. ✗ Project-B

6406535120981. ✗ Project-C

6406535120982. ✗ None of these

Question Number : 59 Question Id : 6406531529721 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Assume there are two periods and prices of consumption good are the same (and equal to 1) in both periods. You earn Rs 10,000 in period 1 and Rs 12,000 in period 2. In each of the two periods, your spending is equal to earnings in that period. The slope of intertemporal budget constraint is -1.12 , then what is the prevailing interest rate?

Options :

6406535120983. ✗ 10%

6406535120984. ✓ 12%

6406535120985. ✗ 15%

6406535120986. ✗ Insufficient information

Question Number : 60 Question Id : 6406531529722 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the present value of an annuity that pays Rs 1,000 for next 25 years (beginning next year) and faces a cost of capital of 5%? (Nearest integer value)

Options :

6406535120987. ✖ Rs 25,000

6406535120988. ✖ Rs 18,394

6406535120989. ✔ Rs 14,094

6406535120990. ✖ Rs 12,500

Question Number : 61 Question Id : 6406531529723 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A typical consumer basket in the economy of Pangia Island consists of 1 unit of clothes, 3 units of food and 2 units of fuel. In 2022, the price of clothes was Rs 200 per unit, the price of food was Rs 50 per unit and the price of fuel was Rs 75 per unit. In 2023, the price of clothes was Rs 180 per unit, the price of food was Rs 70 per unit and the price of fuel was Rs 80 per unit. In 2024, the price of clothes was Rs 220 per unit, the price of food was Rs 80 per unit and the price of fuel was Rs 100 per unit. Considering 2022 as base year, what was the CPI in 2024? (Note: CPI in base year =100)

Options :

6406535120991. ✖ 100

6406535120992. ✖ 110

6406535120993. ✖ 120

6406535120994. ✔ 132

Question Number : 62 Question Id : 6406531529724 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A typical consumer basket in the economy of Pangia Island consists of 1 unit of clothes, 3 units of food and 2 units of fuel. In 2022, the price of clothes was Rs 200 per unit, the price of food was Rs 50 per unit and the price of fuel was Rs 75 per unit. In 2023, the price of clothes was Rs 180 per unit, the price of food was Rs 70 per unit and the price of fuel was Rs 80 per unit. In 2024, the price of clothes was Rs 220 per unit, the price of food was Rs 80 per unit and the price of fuel was Rs 100 per unit. Considering 2022 as base year, what was the annual inflation rate in 2024 with respect to 2023?

Options :

6406535120995. ✖ 10%

6406535120996. ✖ 16%

6406535120997. ✔ 20%

6406535120998. ✖ 32%

Question Number : 63 Question Id : 6406531529725 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Manish runs a private school in his town. He wants to start a bus service for the students. He goes to the bank and asks for a loan of Rs 8,00,000. The bank official tells him that the loan will have a monthly interest rate of 3% for 10 years. What would be the equated monthly instalment (EMI) that he will be paying each month for the next 10 years, beginning next month (Round off to nearest integer)?

Options :

6406535120999. ✖ Rs 12,000

6406535121000. ✖ Rs 22,746

6406535121001. ✔ Rs 24,711

6406535121002. ✖ Rs 28,437

Question Number : 64 Question Id : 6406531529726 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a 20-year zero-coupon bond that promises to pay 1000 units after 20 years. The annualized interest rate is 2.5%. Suppose due to geopolitical tensions, the prevailing interest rate increases by 10 basis points to 2.6%. What would be the change in the value of the bond?

Options :

6406535121003. ✖ increases by 1.93%

6406535121004. ✔ decreases by 1.93%

6406535121005. ✖ increases by 1.20%

6406535121006. ✖ decreases by 1.20%

Question Number : 65 Question Id : 6406531529727 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

On investigating the Indian yield curve, you find that annualized rate on 1-year bond (expiring in September 2026) is 4% and the annualized rate on 2-year bond (expiring in September 2027) is 4.5%. What would be the 1-year interest rate from September 2026 to September 2027?

Options :

6406535121007. ✖ 4.28%

6406535121008. ✖ 4.50%

6406535121009. ✖ 4.72%

6406535121010. ✔ 5.00%

Question Number : 66 Question Id : 6406531529728 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

In the economy of Moldova, the nominal GDP is accounted as 15,000 units and the real GDP is accounted as 12,000 units in 2024, then what is the value of GDP deflator in 2024 for Moldova?